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REGULAR CONTRIBUTION

TRINITY-IDS: a tri-view hybrid architecture with adaptive drift-evidential learning for robust and continual intrusion detection

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Abstract

The rapid evolution of cyber threats in modern network environments demands intelligent intrusion detection systems. This system should be capable of handling complex attack patterns and heterogeneous data distributions. Existing intrusion detection approaches mainly focus on statistical or temporal flow characteristics. This approach neglects the topological relationships among network entities and the uncertainty associated with detection decisions. In this work, a Tri-View Hybrid Architecture with Adaptive Drift-Evidential Learning (TRINITY-IDS) is proposed for robust intrusion detection. Initially, a Density-Aware Hybrid Resampling strategy based on Cluster-SMOTE and Edited Nearest Neighbor (ENN) cleaning is applied to overcome class imbalance and improve minority attack representation. An adversarial feature selection mechanism with gradient reversal learning is used to reduce dataset-specific feature leakage and enhance cross-domain generalization. The extracted features are represented through three complementary views: a statistical view using convolutional feature extraction with channel attention, a temporal view using speciality-token-augmented BiGRU with self-attention for sequential behavior modeling, and a topological view using Graph Attention Networks (GAT) to capture network relationship patterns. A dynamic cross-view adaptive fusion mechanism automatically learns the contribution of each representation for individual samples. In addition, a Dirichlet evidential learning head is used for uncertainty-aware attack classification. It estimates belief and uncertainty values rather than only deterministic predictions. To support evolving network conditions, an online Page-Hinkley drift detector and experience replay mechanism are incorporated for continual adaptation during streaming intrusion detection. Experimental evaluation demonstrates that the proposed TRINITY-IDS achieves an overall accuracy of 98.36%, precision of 98.26%, recall of 98.32%, and F1-score of 98.29%.

Keywords Intrusion detection system (IDS) · Network security · Tri-view learning · Graph Attention Network (GAT) · Bidirectional Gated Recurrent Unit (BiGRU) · Cross-view adaptive fusion · Dirichlet evidential learning

1 Introduction

The rapid growth of digital communication has significantly increased the dependency on cyberspace for critical applications [1]. Cloud computing and Internet of Things (IoT) technologies provide data exchange and intelligent services. This creates new opportunities for cybercriminals to exploit vulnerabilities in network systems. The modern cyber-attacks like denial-of-service (DoS), distributed denial-of-service (DDoS), malware intrusion, ransomware, botnet activities, probing attacks, and zero-day attacks continue to evolve with increasing complexity [2, 3]. These sophisticated threats can compromise the confidentiality and availability of network resources. Therefore, there is a need to develop intelligent and adaptive cybersecurity solutions to detect malicious activities in real-time.

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Intrusion Detection Systems (IDSs) contribute a significant role in cybersecurity [4]. This system continuously monitors network activities and identifies abnormal behaviors. The traditional IDS approaches are mainly based on rule-based and signature-based detection mechanisms where predefined attack patterns are matched against incoming traffic. But, they suffer from limited adaptability against unknown and rapidly evolving threats. Moreover, the increasing volume and complexity of network traffic make conventional detection techniques inefficient for modern heterogeneous environments. Consequently, intelligent IDS solutions based on data-driven approaches have gained significant attention to increase detection accuracy and adaptability.

Recently, Machine Learning (ML) techniques [5] have been adopted for intrusion detection due to their capability to learn discriminative patterns from network traffic data. The conventional ML algorithms like Support Vector Machine (SVM), Decision Tree, Random Forest, and k-Nearest Neighbor have shown promising performance in attack classification. However, these approaches generally depend on handcrafted feature extraction and may fail to capture complex nonlinear relationships present in large-scale network data. Conversely, Deep Learning (DL) [6] approaches have emerged as effective solutions for automatic feature learning and intelligent threat detection. Deep neural like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Transformer models have proved improved capability in extracting spatial and temporal patterns from network traffic. The CNN-based approaches are effective in learning local statistical characteristics. The Transformer-based models improve long-range dependency learning through attention mechanisms. However, existing DL-based IDS approaches still face several challenges like insufficient handling of class imbalance and dependency on dataset-specific features. Also, they suffer from limited cross-domain generalization and lack of uncertainty estimation.

To address these challenges, A Hybrid Multi-View Intrusion Detection System with Adaptive Drift-Evidential Learning called TRINITY-IDS is proposed for intruder detection. The major contributions of this work are summarized as follows:

1. A novel TRINITY-IDS hybrid multi-view intrusion detection model is proposed. It integrates statistical, temporal, and topological feature learning for complete cyber-threat analysis.
2. A Density-Aware Hybrid Resampling and Adversarial-Validated Feature Selection strategy is developed to address class imbalance and improve domain-independent feature representation.
3. A multi-view representation learning architecture is applied
4. A Cross-View Adaptive Fusion mechanism is designed to dynamically combine heterogeneous feature representations based on sample-specific importance.
5. A Dirichlet Evidential Learning-based classification approach is incorporated to provide reliable intrusion prediction with uncertainty estimation.
6. An Online Concept Drift Detection and Continual Experience Replay mechanism is developed for adaptive learning against evolving cyber threats.

The novelty of TRINITY-IDS does not arise from the individual use of CNN, BiGRU, GAT, or attention mechanisms. It is achieved through their unified integration with adversarial feature selection, adaptive cross-view fusion, evidential uncertainty estimation, and continual drift adaptation. Compared to conventional hybrid IDS models, TRINITY-IDS jointly captures statistical, temporal, and topological traffic characteristics. It dynamically weights their contributions, quantifies prediction uncertainty, and adapts to evolving traffic patterns through drift detection and experience replay.

2 Related work

Various researchers have explored feature learning and intelligent optimization strategies to enhance the robustness of IDS models. Zhendong Wang et al. [7] proposed an integrated deep IDS based on Stacked Denoising Autoencoder- Extreme Learning Machine (SDAE-ELM) for network intrusion detection. In their study, Yakubu Imrana et al. [8] proposed a Bidirectional Long Short-Term Memory (BiDLSTM)-based intrusion detection system for network security applications. The approach was validated using the NSL-KDD dataset and achieved improved accuracy.

MD Moizuddin et al. [9] suggested a two-stage intrusion detection model. The important traffic features are selected using Generalized Mean Grey Wolf Optimization approach. Also, an ElasticNet Contractive Autoencoder is applied to extract effective attack representations from network data. Likewise, Muaadh A et al. [10] developed a Sparse Autoencoder combined CNN for attack classification. Here, the Sparse Autoencoder is used for dimensionality reduction.

A three-stage model is proposed by Fu Y et al. [11] for network intrusion detection. This work combines CNN with an attention strategy for spatial learning. Then, Bi-LSTM is applied to extract spatial and temporal characteristics of network traffic. Similarly, V. Hnamte et al. [12] proposed a hybrid LSTM-Autoencoder model for network attack detection. The work developed an encoder-based feature learning mechanism combined with LSTM sequence modeling.

In their work, Adefemi K.O. et al. [13] proposed a CNN-GRU-based intrusion detection model for IoT environments. The work integrated CNN layers for spatial feature extraction

and GRU layers for temporal dependency learning. Similarly, Afraji D.M.A.A. et al. [14] proposed a hybrid CNN-LSTM-GRU architecture for intrusion detection. Elnakib O et al. [15] recommended an Enhanced Intrusion Detection Deep Learning Model (EIDM) for multi-class attack detection.

An explainable XGRU-IDS model is developed by Jawad Ahmad et al. [16] for intrusion detection in industrial IoT networks. The work integrated Extra Trees Classifier for feature selection and GRU for sequential attack analysis. Mohammad Mehedi Hassan et al. [17] proposed a hybrid CNN- Weight-Dropped LSTM (WDLSTM) based intrusion detection model for network security. The work uses a weight-dropped LSTM for learning long-term dependencies.

Selman Hizal et al. [18] proposed a two-level intrusion detection framework for identifying DDoS attacks in IoT networks. The work developed binary and multi-class classification models using fully connected, CNN, and LSTM-based deep learning architectures. The study explored different deep learning structures for detecting attack categories and subcategories.

Vanlalruata Hnamte et al. [19] suggested a lightweight deep CNN-based intrusion detection model. The work developed an optimized DCNN for feature extraction. Likewise, Edosa Osa et al. [20] proposed a Deep Neural Network-based intrusion detection model for computer networks. This work applied data balancing techniques of SMOTE and random sampling to improve attack data representation.

Shaikh JA et al. [21] proposed a DL-based intrusion detection model combined with Random Forest and LSM techniques. This approach detects important network attributes for better model learning.

Fatimah Alhayan et al. [22] proposed an Enhanced Intrusion Detection in Cybersecurity using Ensemble Learning with Improved Beluga Whale Optimization (IDCS-ELIBWO). The study explored an improved IBWO to enhance intrusion detection capability.

S.M. Hosseini et al. [23] proposed attention-based CNN-GRU intrusion detection architectures for cybersecurity applications. In their study, Wang H. et al. [24] proposed a hybrid model based on a Convolutional Autoencoder and Temporal Convolutional Network. In addition, this work uses incremental learning strategies based on weight alignment and task-specific feature sharing.

Waqas Ishtiaq et al. [25] suggested a dual attention-based deep learning model for IoT intrusion detection. This work integrates multi-scale CNN and BiGRU to extract important attack features. Hussein S.A. et al. [26] proposed a hybrid AE-GMM-XGBoost (Autoencoder–Gaussian Mixture Model–Extreme Gradient Boosting) model for intrusion detection. It involves GMM-based probabilistic feature analysis and XGBoost-based supervised classification.

In their study, Dash N et al. [27] proposed an optimized LSTM-based intrusion detection model using metaheuristic optimization algorithms. The hyperparameters of the LSTM are tuned using the Salp Swarm Algorithm. Similarly, Farhan M et al. [28] proposed a Sequential DNN-based intrusion detection model integrated with Extra Tree feature selection.

Alabdulatif A et al. [29] proposed an explainable ensemble learning system for real-time intrusion detection. It combines ANN and SVM classifiers with Random Forest-based prediction fusion. Md. Alamgir Hossain et al. [30] suggested a Deep Q-Learning-based Intrusion Detection System (DQ-IDS) using reinforcement learning. The work developed a Deep Q-Network model with experience replay and adaptive exploration strategies.

3 TRINITY-IDS

The TRINITY-IDS uses a tri-view hybrid intrusion detection architecture for effective cyber threat identification, as shown in Fig. 1. Initially, a density-aware hybrid resampling technique based on Cluster-SMOTE and ENN is applied. This step is used to balance attack categories and to improve minority class learning. Then, an adversarial feature selection mechanism is used to remove dataset-specific biases and enhance generalization across different network environments. The extracted features are analyzed through three complementary modules: a CNN-based statistical view with channel attention, a BiGRU-based temporal view with self-attention and a Graph Attention Network-based topological view for capturing network interactions. These multi-view features are dynamically combined using adaptive fusion. A Dirichlet evidential learning module is applied for uncertainty-aware attack prediction. In addition, an online drift detection and experience replay mechanism is used for continuous adaptation to change network behaviours (Fig. 2).

3.1 density-aware hybrid resampling module (DAHR)

In real-world network traffic, normal connections constitute the majority of samples.

In contrast, several attack categories contain only a limited number of instances.

This imbalance creates challenges for accurate detection of minority attack classes.To address this issue, the proposed module integrates Cluster-SMOTE with ENN cleaning. The Cluster-SMOTE mechanism generates synthetic attack samples based on the local distribution of minority classes. The ENN technique removes noisy and ambiguous samples near

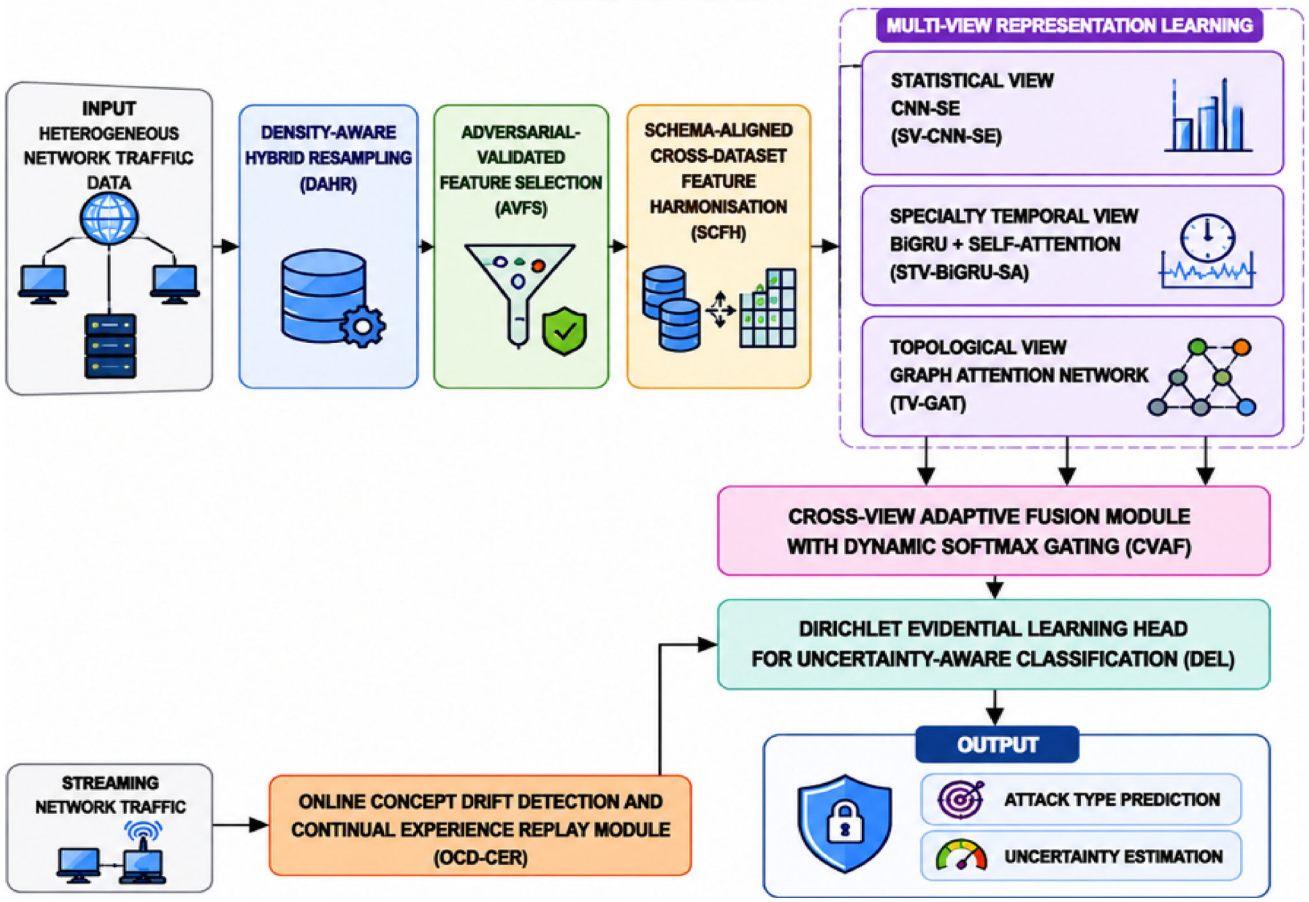


Fig. 1 TRINITY-IDS

class boundaries. Let the original intrusion dataset be represented as:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

where x_i represents the network traffic feature vector and y_i denotes the corresponding class label. The minority attack samples belonging to class c are extracted as:

$$X_c = \{x_i | y_i = c\} \tag{2}$$

To preserve the inherent distribution of different attack behaviors, the minority samples are divided into multiple clusters using K-means clustering. The clustering objective is formulated as:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \tag{3}$$

where C_k represents the kth cluster and μ_k denotes the cluster centroid. This clustering process generates synthetic samples within similar attack behavior regions.

It avoids random sample generation across the entire feature space. For each minority sample x_i, its nearest neighbor x_j within the same cluster is selected. Then, a synthetic sample is generated using linear interpolation:

$$x_{\text{new}} = x_i + \lambda\left(x_j - x_i\right) \tag{4}$$

where λ is a random interpolation factor ranging between 0 and 1 ($\lambda \in [0,1]$). The generated samples increase the representation of minority attack categories and maintain their original characteristics. After synthetic sample generation, ENN-based cleaning is applied to remove incorrectly generated or overlapping samples. For each sample, the neighborhood labels are analyzed, and the majority class among its nearest neighbors is determined as:

$$y_{\text{maj}} = \text{Mode}(y_1, y_2, \ldots, y_k) \tag{5}$$

If the original class label of the sample differs from the dominant neighborhood class, the sample is considered noisy and removed:

$$x_i \notin D_{\text{clean}}, \quad \text{if } y_i \neq y_{\text{maj}} \tag{6}$$

The final refined training dataset is obtained as:

$$D' = \{(x_i, y_i)\}_{i=1}^{N'} \tag{7}$$

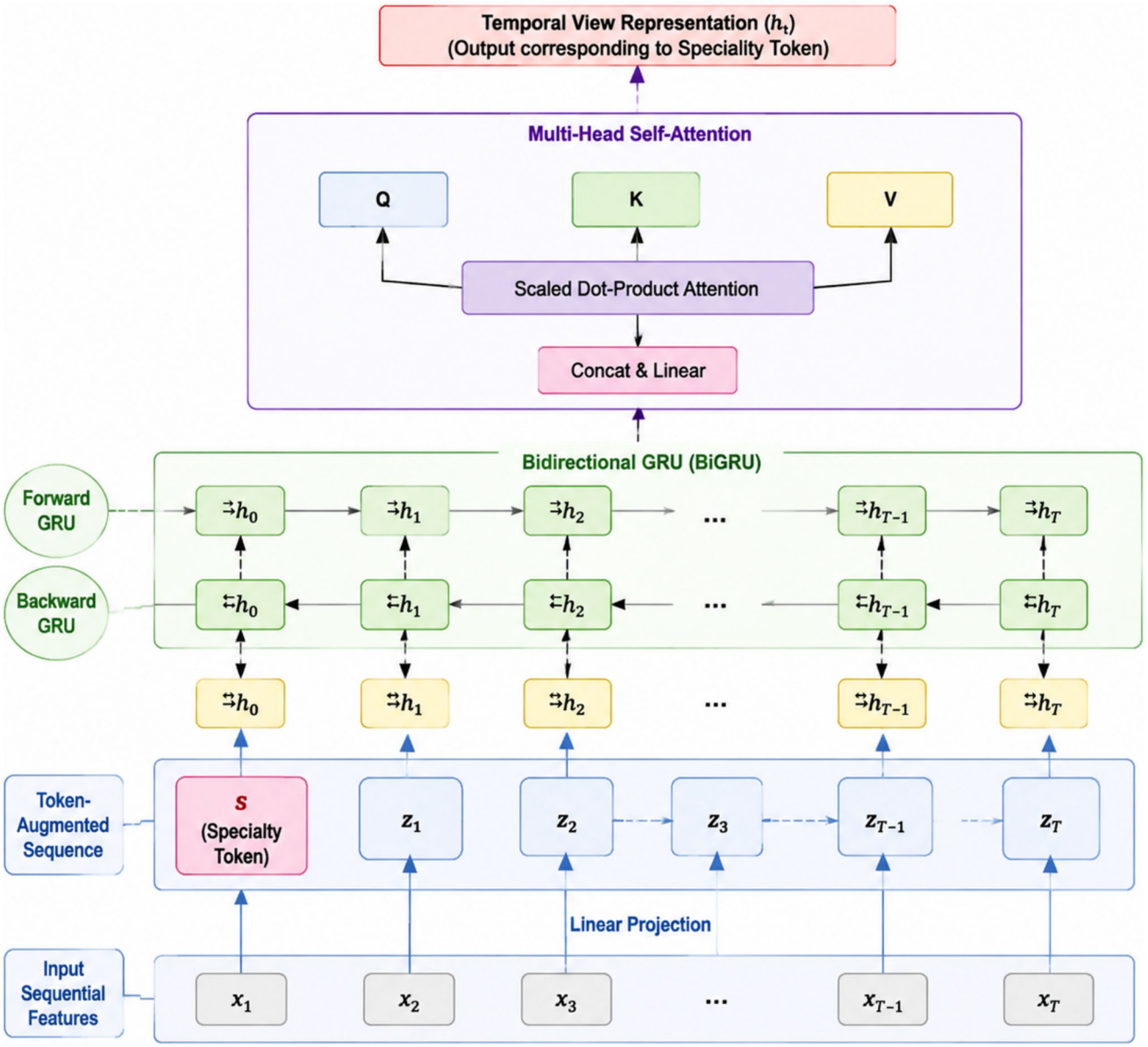


Fig. 2 STV-BiGRU-SA architecture

where $N\prime$ represents the number of samples after balancing and noise elimination. The DAHR module reduces class bias and achieves a balanced input distribution for subsequent feature extraction and classification.

3.2 Adversarial-validated feature selection module (AVFS)

An AVFS module is used to eliminate dataset-specific feature biases and improve the generalization capability of the intrusion detection model. In multi-dataset intrusion detection environments, different datasets may contain unique feature distributions or hidden signatures related to their collection sources. These dataset-specific characteristics can negatively influence the learning process and reduce the robustness of the model when applied to unseen network environments. To overcome this limitation, the proposed module utilizes an adversarial learning strategy based on a Gradient Reversal Layer (GRL). Here, a domain discriminator is trained to identify dataset-specific information and the feature extractor is optimized to suppress such information. Let the extracted feature representation from the resampled dataset be represented as:

$$X = \{x_1, x_2, \ldots, x_N\} \tag{8}$$

where $x_i \in \mathbb{R}^d$ denotes the ith traffic feature vector with d feature dimensions. Each sample is associated with a domain

label:

$$d_i \in \{1, 2, \ldots, M\} \tag{9}$$

where M represents the number of different datasets or network environments.The AVFS module uses a domain discriminator network $D(\cdot)$ to identify the source domain of the input features. The discriminator is formulated as:

$$D(X) = \text{Softmax}(W_d X + b_d) \tag{10}$$

where W_d and b_d represent the discriminator weight and bias parameters, respectively. To prevent the extracted features from retaining domain-specific information, a GRL is added between the feature representation and the discriminator. During forward propagation, the GRL maintains the original feature representation:

$$\text{GRL}(x) = x \tag{11}$$

However, during backpropagation, the gradient is reversed and scaled by a factor α:

$$\frac{\partial \text{GRL}(x)}{\partial x} = -\alpha I \tag{12}$$

where α controls the adversarial learning strength and I represents the identity matrix. This reversal forces the feature extraction process to learn domain-invariant representations and confuses the domain discriminator. The discriminator optimization objective is defined using cross-entropy loss:

$$L_D = -\sum_{i=1}^{N} d_i \log(D(x_i))) \tag{13}$$

where d_i represents the actual dataset domain label and $D(x_i)$ denotes the predicted domain probability. The feature extractor aims to minimize the intrusion classification error and maximizing the confusion of the domain discriminator. Therefore, the overall adversarial optimization objective is expressed as:

$$L_{\text{AVFS}} = L_{\text{IDS}} - \lambda L_D \tag{14}$$

where, L_{IDS} represents the intrusion classification loss, L_D represents the domain discrimination loss and λ represents the adversarial balancing coefficient. After adversarial training, the importance of each feature is calculated based on the discriminator gradient contribution:

$$S_j = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{\partial D(x_i)}{\partial x_{ij}} \right| \tag{15}$$

where S_j represents the importance score of the jth feature. Features with higher dataset leakage scores are removed, and the remaining features are selected as:

$$F_{\text{selected}} = \left\{ f_j \middle| S_j < S_{\text{threshold}} \right\} \tag{16}$$

The selected feature representation contains reduced dataset-specific information and improved attack-related characteristics. Thus, the proposed AVFS module enhances cross-dataset adaptability and reduces feature leakage.

3.3 Schema-aligned cross-dataset feature harmonisation module (SCFH)

A SCFH module is used to achieve effective feature alignment across heterogeneous intrusion detection datasets. In practical network security scenarios, the data collected from different sources may contain varying feature names and feature dimensions. These inconsistencies reduce the effectiveness of conventional learning models when multiple datasets are combined for generalized intrusion detection. The SCFH uses a unified feature representation through semantic feature embedding and adaptive projection. The module maps heterogeneous feature schemas into a common latent representation space and preserves the discriminative characteristics required for attack classification. Let the input network traffic sample be represented as:

$$X = \{x_1, x_2, ..., x_m\} \tag{17}$$

where m denotes the number of input attributes. Since different datasets contain different feature structures, each feature column is assigned a unique feature identity index:

$$C = \{c_1, c_2, \ldots, c_m\} \tag{18}$$

An embedding layer is employed to convert the feature identities into dense semantic representations:

$$E_c = \phi(c_i) \tag{19}$$

where E_c represents the embedding vector of the ith feature and $\phi(\cdot)$ denotes the learnable embedding function. The SCFH module uses a column embedding matrix to capture the semantic relationship between heterogeneous feature attributes:

$$Z_e = W_e C + b_e \tag{20}$$

where, W_e represents the feature embedding weight matrix, b_e represents the bias term and Z_e denotes the embedded feature representation. The embedded feature vectors are combined with their corresponding feature values to preserve

both semantic information and numerical characteristics. The weighted feature representation is obtained as:

$$H_i = E_i \times x_i \tag{21}$$

where E_i represents the embedding vector of the ith feature and x_i represents its corresponding traffic value. The combined feature representation is aggregated and projected into a common latent space using a nonlinear transformation:

$$H_{\text{SCFH}} = \text{ReLU}\big(W_p H + b_p\big) \tag{22}$$

where W_p and b_p represent the projection parameters. The resulting harmonized representation can be expressed as:

$$Z_{\text{SCFH}} = H_{\text{SCFH}} \in \mathbb{R}^d \tag{23}$$

where d denotes the unified feature dimension.

3.4 Statistical view with CNN and squeeze-and-excitation channel attention module (SV-CNN-SE)

The intrusion traffic contains complex behavioral patterns.Each attack category has distinct characteristics in network behavior.These variations are reflected in flow-level attributes like packet distribution, connection properties, and feature correlations.To effectively capture these characteristics, the proposed module uses a one-dimensional CNN (1D-CNN) integrated with a Squeeze-and-Excitation (SE) channel attention mechanism. The CNN component learns local feature patterns. The SE mechanism adaptively highlights important feature channels and suppresses less informative representations. Let the harmonized feature representation obtained from the SCFH module be represented as:

$$Z_{\text{SCFH}} = [z_1, z_2, \ldots, z_d] \tag{24}$$

where d represents the number of aligned features. The input feature vector is reshaped into a one-dimensional feature sequence:

$$X_s \in \mathbb{R}^{1\times d} \tag{25}$$

The statistical representation is extracted using multiple convolution kernels with different receptive fields. The convolution operation is formulated as:

$$H_k = \sigma(W_k * X_s + b_k) \tag{26}$$

where, W_k represents the convolution kernel, b_k denotes the bias parameter, $*$ represents the convolution operation and $\sigma(\cdot)$ represents the nonlinear activation function. To capture statistical patterns with different local dependencies, multiple convolution filters are applied:

$$H = H_3 + H_5 \tag{27}$$

where H_3 and H_5 represent the feature maps generated using kernel sizes 3 and 5, respectively. This multi-scale convolution is used for the model to learn both short-range and extended feature interactions from network traffic data. After convolution-based feature extraction, a Squeeze-and-Excitation attention mechanism is applied to recalibrate the learned feature channels. The squeeze operation performs global average pooling to generate channel-wise statistics:

$$s_c = \frac{1}{L}\sum_{i=1}^{L} H_c(i) \tag{28}$$

where, s_c denotes the channel descriptor, L represents the feature map length and $H_c(i)$ represents the activation value of channel c. The excitation operation learns adaptive channel importance weights using two fully connected layers:

$$e = \sigma(W_2\delta(W_1 s)) \tag{29}$$

where, W_1 and W_2 are learnable parameters, $\delta(\cdot)$ represents the ReLU activation and $\sigma(\cdot)$ represents the sigmoid function. The recalibrated statistical representation is obtained by:

$$\hat{H}_c = e_c \times H_c \tag{30}$$

where e_c represents the learned attention coefficient for the cth feature channel. Finally, the statistical feature representation is generated through channel-wise aggregation:

$$Z_s = \frac{1}{C}\sum_{c=1}^{C} \hat{H}_c \tag{31}$$

where C represents the total number of feature channels.

3.5 Speciality temporal view with token-augmented BiGRU and self-attention module (STV-BiGRU-SA)

The Temporal View module is used to capture the sequential dependencies and evolving behavioral patterns present in network traffic streams. The conventional recurrent models focus only on sequential dependencies and may fail to identify important attack-related temporal regions. To overcome this limitation, the proposed module integrates a learnable speciality token, a Bidirectional Gated Recurrent Unit (BiGRU), and a Multi-Head Self-Attention mechanism

to enhance temporal representation learning. The speciality token acts as an adaptive global representation that aggregates important attack-related temporal information from the entire sequence. Let the sequential feature representation obtained from the harmonization module be represented as:

$$X_t = \{x_1, x_2, \ldots, x_T\} \tag{32}$$

where T represents the sequence length and x_t denotes the traffic feature vector at time step t. Initially, the input sequence is transformed into a hidden feature space using a linear projection:

$$h_t = W_f x_t + b_f \tag{33}$$

where W_f and b_f represent the projection weight and bias parameters, respectively. To enhance global temporal understanding, a learnable special token T_s is used and appended before the input sequence:

$$X'_t = [T_s, h_1, h_2, \ldots, h_T] \tag{34}$$

where T_s represents the trainable token embedding to learn important temporal characteristics related to intrusion evidence. This token-based representation is used for the model to capture global attack-specific patterns from sequential traffic behavior. The token-augmented sequence is processed using a Bidirectional GRU network to learn forward and backward temporal dependencies. The forward and backwards hidden states are computed as:

$$\overrightarrow{h_t} = \mathrm{GRU}_f\left(x_t, \overrightarrow{h_{t-1}}\right) \tag{35}$$

$$\overleftarrow{h_t} = \mathrm{GRU}_b\left(x_t, \overleftarrow{h_{t+1}}\right) \tag{36}$$

The bidirectional temporal representation is obtained by combining both directional states:

$$H_t = \left[\overrightarrow{h_t}\,;\ \overleftarrow{h_t}\right] \tag{37}$$

Where; represents the concatenation operation. The BiGRU structure is used for the model to learn both previous and future contextual relationships in network traffic sequences. The BiGRU captures sequential dependencies. But, all temporal positions may not contribute equally to intrusion identification. Therefore, a self-attention mechanism is applied to assign adaptive importance weights to different sequence elements. The query, key, and value matrices are generated as:

$$Q = HW_Q, \quad K = HW_K, \quad V = HW_V \tag{38}$$

where W_Q, W_K, and W_V are learnable projection matrices. The attention score is calculated as:

$$\mathrm{Attention}(Q,\ K,\ V) = \mathrm{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{39}$$

where d_k represents the dimensionality of the key vectors. The attention mechanism is used to focus on critical temporal regions associated with attack behaviors. The output corresponding to the speciality token is extracted as the final temporal representation:

$$Z_t = W_o H_{\text{token}} + b_o \tag{40}$$

where H_{token} represents the hidden state associated with the speciality token and Z_t denotes the final temporal feature representation.

3.6 Topological view using Graph Attention Network module (TV-GAT)

The Topological View module is used to model the non-Euclidean relationships among network traffic features using a GAT. In conventional intrusion detection approaches, the network traffic samples are generally represented as independent feature vectors. This limits the ability to capture complex interactions among different network entities and communication patterns. To capture these hidden structural dependencies, the proposed module transforms network traffic information into a graph-based representation and applies graph attention learning to extract topology-aware features. The network traffic graph is represented as:

$$G = (V,\ E) \tag{41}$$

where V denotes the set of nodes representing traffic entities or feature components, and E represents the connections between nodes. Each node contains an initial feature representation:

$$X_v = \{x_1, x_2, \ldots, x_n\} \tag{42}$$

where x_v represents the feature vector of node v. The relationship between connected nodes is represented using an adjacency matrix:

$$A = \left[a_{ij}\right] \tag{43}$$

where:

$$a_{ij} = \begin{cases} 1, & \text{if nodes } i \text{ and } j \text{ are connected} \\ 0, & \text{otherwise} \end{cases}$$

The proposed module applies Graph Attention Network layers to learn the importance of neighboring nodes. Initially, node features are transformed using a learnable weight matrix:

$$h_i = W x_i \tag{44}$$

where W represents the trainable graph transformation matrix. For each connected node pair $(i,\ j)$, the attention coefficient is calculated as:

$$e_{ij} = \text{LeakyReLU}\Big(\text{a}^T \big[h_i \| h_j\big]\Big) \tag{45}$$

where, a represents the attention weight vector and $\|$ denotes feature concatenation. The normalized attention coefficient is obtained using the Softmax operation:

$$\alpha_{ij} = \frac{\exp\big(e_{ij}\big)}{\sum_{k\in\mathcal{N}_i} \exp(e_{ik})} \tag{46}$$

where $\mathcal{N}_i$ represents the neighboring nodes of node i. These attention weights are used to assign higher importance to nodes that contribute more significant information for intrusion identification. The updated node representation is computed by aggregating the weighted neighborhood information:

$$h_i' = \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij} W h_j\right) \tag{47}$$

where $\sigma(\cdot)$ represents the nonlinear activation function. To enhance feature learning capability, multiple attention heads are used. Each attention head independently learns different relational patterns:

$$h_i' = \Big\|_{m=1}^{M} \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}^m W^m h_j\right) \tag{48}$$

where M represents the number of attention heads. After graph-based feature propagation, a graph-level representation is obtained through global mean pooling:

$$Z_g = \frac{1}{N}\sum_{i=1}^{N} h_i' \tag{49}$$

where Z_g represents the final topological feature representation.

3.7 Cross-view adaptive fusion module with dynamic softmax Gating (CVAF)

The CVAF module combines complementary information obtained from statistical, temporal, and topological representations. This module dynamically assigns importance weights to each feature view according to the characteristics of individual input samples. The CVAF module uses a sample-level softmax gating mechanism to learn adaptive contributions from the three feature representations. Let the extracted representations from the three learning views be represented as:

$$Z_s \in \mathbb{R}^d,\ Z_t \in \mathbb{R}^d,\ Z_g \in \mathbb{R}^d \tag{50}$$

where Z_s, Z_t, and Z_g denote the statistical, temporal, and topological feature representations, respectively. Initially, the three representations are concatenated to form a combined feature vector:

$$Z_c = \big[\, Z_s \| Z_t \| Z_g \big] \tag{51}$$

where $\|$ represents the concatenation operation. The concatenated representation is provided to a gating network to estimate the contribution of each feature view. The gating function is formulated as:

$$G = \text{Softmax}\big(W_g Z_c + b_g\big) \tag{52}$$

where, W_g represents the learnable gating weight matrix, b_g represents the bias parameter and G represents the learned view importance vector. The gating vector consists of three adaptive coefficients:

$$G = \big[\alpha_s,\ \alpha_t,\ \alpha_g\big] \tag{53}$$

where, α_s represents the importance weight of the statistical view, α_t represents the importance weight of the temporal view and α_g represents the importance weight of the topological view. The final fused representation is generated by weighted aggregation of the three feature views:

$$Z_f = \alpha_s Z_s + \alpha_t Z_t + \alpha_g Z_g \tag{54}$$

where Z_f represents the adaptive fused feature representation.

3.8 Dirichlet evidential learning head for uncertainty-aware classification (DEL)

The DEL Head is used to perform uncertainty-aware intrusion classification. This module estimates both classification belief and predictive uncertainty by modelling the output

distribution using a Dirichlet distribution. This supports TRINITY-IDS in distinguishing between confident predictions and uncertain detection decisions. Let the adaptive fused representation obtained from the Cross-View Adaptive Fusion module be represented as:

$$Z_f \in \mathbb{R}^d \tag{55}$$

where d represents the dimension of the fused feature space. The fused representation is passed through a fully connected layer to generate class-wise evidence values:

$$e_i = W_e Z_f + b_e \tag{56}$$

where, W_e represents the classification weight matrix, b_e represents the bias parameter and e_i denotes the evidence value corresponding to the ith intrusion class.

To ensure positive evidence values, the Softplus activation function is applied:

$$e_i = \text{Softplus}(e_i) \tag{57}$$

The obtained evidence values are converted into Dirichlet concentration parameters:

$$\alpha_i = e_i + 1 \tag{58}$$

where α_i represents the Dirichlet parameter for the ith class. The total evidence strength is calculated as:

$$S = \sum_{i=1}^{K} \alpha_i \tag{59}$$

where K denotes the number of intrusion categories. The belief mass assigned to each class is computed as:

$$b_i = \frac{\alpha_i}{S} \tag{60}$$

where b_i represents the confidence belief associated with class i. The uncertainty of the prediction is estimated based on the total available evidence:

$$u = \frac{K}{S} \tag{61}$$

where, u represents the overall predictive uncertainty, K represents the number of classes and S represents the total evidence strength. A higher evidence value produces a larger S which results in lower uncertainty, whereas insufficient evidence increases uncertainty. This strategy is used for the model to identify unreliable predictions instead of forcing incorrect classifications. The training process utilizes an Evidential Deep Learning (EDL) loss function. It combines data fitting and uncertainty regularization. The classification error component is formulated as:

$$L_{\text{err}} = \sum_{i=1}^{K} y_i (\psi(S) - \psi(\alpha_i)) \tag{62}$$

where, y_i represents the one-hot encoded class label and $\psi(\cdot)$ denotes the digamma function. To prevent overconfident predictions, a Kullback–Leibler (KL) divergence regularization term is incorporated:

$$L_{\text{KL}} = D_{\text{KL}}(\text{Dir}(\alpha) + \text{Dir}(1)) \tag{63}$$

The final evidential loss is expressed as:

$$L_{\text{EDL}} = L_{\text{err}} + \lambda L_{\text{KL}} \tag{64}$$

where λ represents the regularization coefficient controlling uncertainty calibration. The DEL Head improves the reliability of TRINITY-IDS by generating the predicted attack category. Also, it provides the confidence level associated with each decision to enhance prediction interpretability and trustworthiness.

3.9 Online concept drift detection and continual experience replay module (OCD-CER)

In this work, the OCD-CER module is applied to maintain detection performance under continuously changing network environments. The proposed module combines a Page-Hinkley drift detection mechanism with an experience replay strategy to identify performance degradation and perform incremental model adaptation. The online detection process continuously monitors the prediction loss generated during streaming inference. Let the loss value obtained from the tth incoming traffic batch be represented as l_t. The mean loss value is updated incrementally as:

$$\mu_t = \mu_{t-1} + \frac{l_t - \mu_{t-1}}{t} \tag{65}$$

where, μ_t represents the estimated mean loss at time t and l_t represents the current prediction loss. The cumulative deviation value is calculated as:

$$M_t = \alpha M_{t-1} + (l_t - \mu_t - \delta) \tag{66}$$

where, M_t represents the accumulated deviation measure, α represents the forgetting factor and δ represents the allowed variation margin. A drift condition is detected when $M_t > \theta$, where θ represents the predefined drift threshold. When this condition is satisfied, the system identifies that the current

traffic distribution has changed and triggers an adaptive learning process.

After detecting concept drift, the proposed model activates an Experience Replay Buffer to preserve previously observed traffic knowledge and avoid catastrophic forgetting. The replay memory is represented as:

$$\mathcal{B} = \{(x_i, y_i)\}_{i=1}^{N_b} \tag{67}$$

where N_b represents the maximum buffer capacity. Incoming samples are continuously stored in the buffer:

$$\mathcal{B}_t = \mathcal{B}_{t-1} \cup \{(x_t, y_t)\} \tag{68}$$

The older samples are removed when the buffer reaches its maximum capacity. During incremental updating, a mini-batch of historical samples is randomly selected from the replay memory:

$$\mathcal{B}_r = \{(x_r, y_r)\}_{r=1}^{m} \tag{69}$$

Then, the model is updated using a combined loss function containing both new streaming samples and replayed historical samples:

$$L_{\text{total}} = L_{\text{new}} + \beta L_{\text{replay}} \tag{70}$$

where, L_{new} represents the loss from current traffic samples, L_{replay} represents the loss from replayed historical samples and β controls the contribution of previous knowledge. The updated model parameters are optimized as:

$$\Theta_{t+1} = \Theta_t - \eta \nabla L_{\text{total}} \tag{71}$$

where, Θ represents the model parameters and η represents the learning rate.

4 Results and discussion

To evaluate the performance of the TRINITY-IDS model, experiments were conducted on a unified multi-dataset benchmark. This dataset is synthesised from three standard network cybersecurity datasets: NSL-KDD, ToN-IoT, and UNSW-NB15. The dataset consists of 1,366,299 total network traffic samples. 80% of the samples are used to train the model. The remaining 20% of the data is used for testing. The model is implemented using PyTorch 2.2 on an NVIDIA RTX 4090 GPU ($24GB$ VRAM). The model parameters are given in Table 1.

The training performance of the proposed TRINITY-IDS model is shown in Fig. 3. The training and validation loss values gradually decrease with increasing epochs. The close convergence between training and validation curves proves stable learning without significant overfitting. Similarly, the classification accuracy increases with increasing epochs. This shows the capability of TRINITY-IDS to learn discriminative intrusion patterns and achieve reliable generalization for complex network traffic classification.

Table 1 TRINITY-IDS parameter settings

Architectural layer / sub-block	Operational parameter setting
KMeans intra-cluster SMOTE	$N_{clusters} = 3$, $K_{neighbors} = 5$
ENN neighborhood cleaning	$K_{enn} = 5$, Majority threshold
Domain discriminator	2-Layer MLP ($64 \rightarrow 3$), ReLU
GRL	Scale factor $\alpha = 1.0$
Column entity embedding	$D_{embed} = 64$, $D_{proj} = 128$
1D-CNN + SE attention	Conv1D ($K = 3,5$), Reduction $r = 4$
Specialty token BiGRU + MHA	$D_{hidden} = 128$, Heads $= 2$, Steps $= 10$
GAT	$K_{GAT1} = 2heads$, $K_{GAT2} = 1head$
Softmax gating network	2-Layer MLP ($384 \rightarrow 128 \rightarrow 3$)
Dirichlet parameterization	Linear Layer ($128 \rightarrow 20$), Softplus
Page-Hinkley monitor	$\delta = 0.005$, $U_{threshold} = 30.0$, $\alpha = 0.999$
Experience replay buffer	Capacity $M = 2000$, Replay Batch $= 64$
AdamW + cosine scheduler	$\eta_0 = 10^{-3}$, $\beta = (0.9,0.999)$, $\lambda = 10^{-4}$

The original merged intrusion dataset contains highly imbalanced attack distributions. The dominant classes contain significantly larger samples when compared with minority attack categories. The raw input distribution shows severe imbalance among different attack classes (Fig. 4). The resampled distribution gets a more balanced representation across attack categories. It prevents the classifier from being biased toward majority classes. Also, it enhances detection capability for low-frequency cyber threats.

The feature validation process analyzes candidate features based on dataset identity leakage probability. The proposed adversarial validation mechanism identifies highly dataset-specific attributes and removes potentially biased features, as shown in Fig. 5. It supports TRINITY-IDS in achieving better adaptability across heterogeneous intrusion datasets.

The cross-dataset feature harmonisation analysis is shown in Fig. 6. Before alignment, different intrusion datasets occupy separated feature distributions due to variations in feature dimensions and extraction strategies. The raw feature visualization shows distinct clusters corresponding to individual datasets. After applying learnable column embedding,

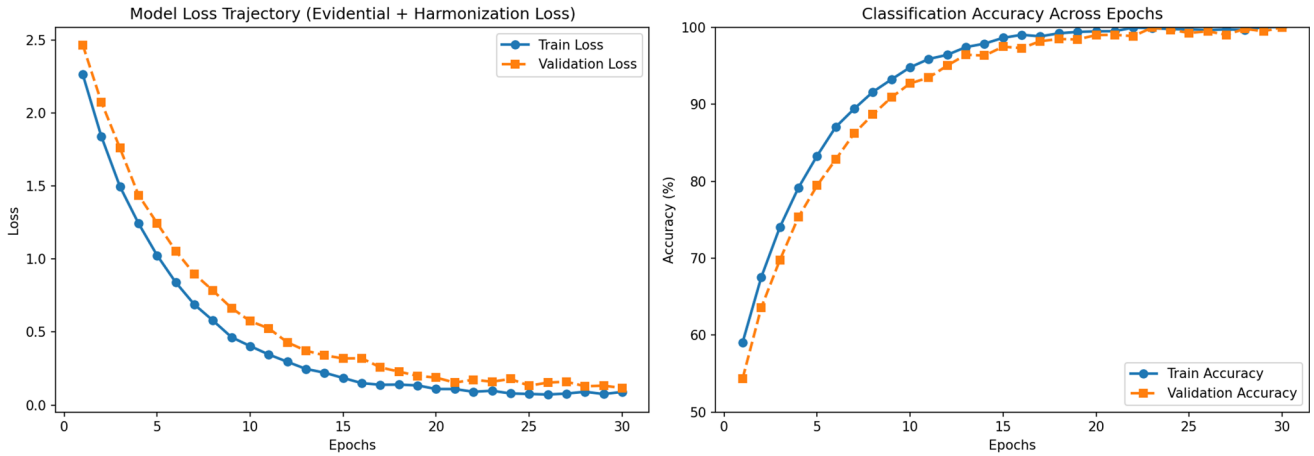


Fig. 3 TRINITY-IDS training validation

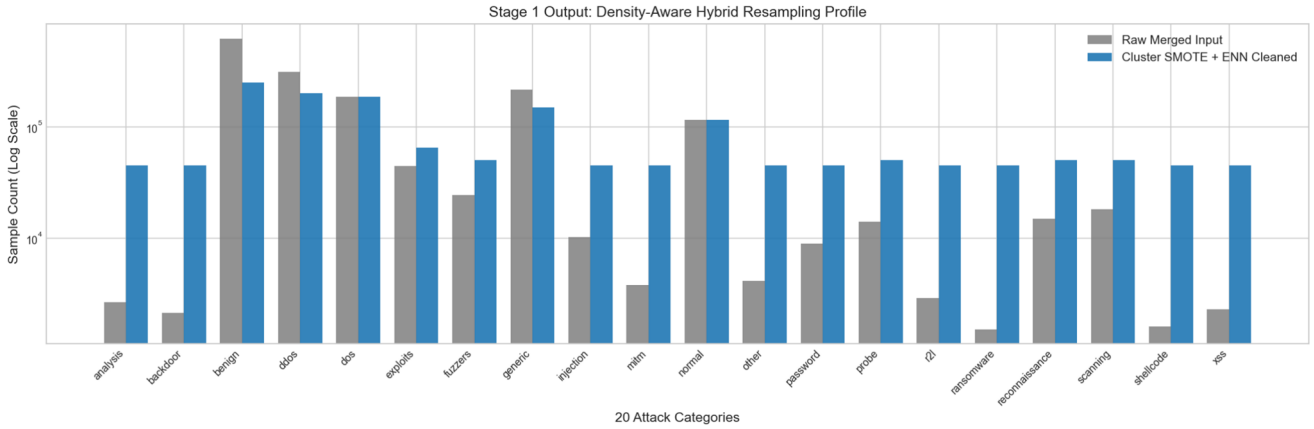


Fig. 4 Density-aware hybrid resampling analysis

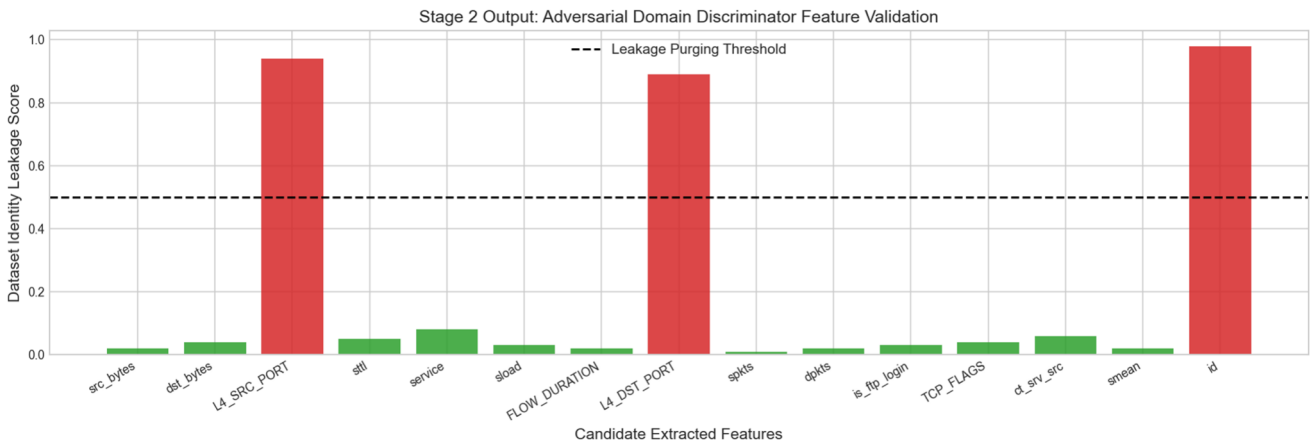


Fig. 5 Adversarial-validated feature selection analysis

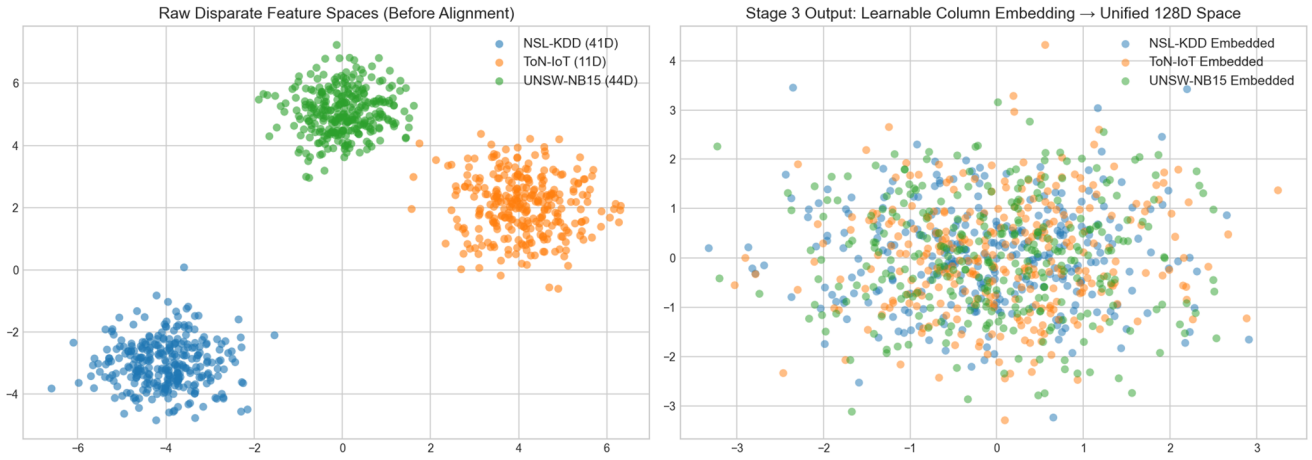


Fig. 6 The cross-dataset feature harmonisation analysis

heterogeneous feature spaces are projected into a unified latent representation space. The aligned embedding distribution shows increased feature overlap among datasets. This confirms that the proposed SCFH module reduces domain discrepancy and maintains unified multi-dataset learning.

The temporal representation learning performance is analyzed using the Speciality Temporal View based on BiGRU with self-attention (Fig. 7). The attention allocation visualization proves that different sequence positions receive different importance values. The higher attention weights are assigned to critical time steps. It is observed that the model successfully identifies significant temporal patterns associated with intrusion behaviors.

The confusion matrix obtained for the model is shown in Fig. 8. The diagonal dominance in the confusion matrix indicates that most samples are correctly classified into their corresponding attack classes. The model achieves high recognition capability for both frequent and minority attack categories. The low off-diagonal misclassification values indicate that the proposed model recognizes between similar attack categories. The class-wise precision, recall, and F1-score performance of the TRINITY-IDS model is given in Table 2. The model achieves consistently high performance across all classes with F1-scores above 0.9680. It proves the effective recognition of both normal and attack traffic patterns. The highest performance is observed for the analysis, benign, DDoS, and generic classes. The balanced precision and recall values confirm that TRINITY-IDS effectively reduces false detections and maintains reliable detection of different cyber-attacks.

The performance of the TRINITY-IDS is compared with existing intrusion detection approaches. The evaluated results are given in Table 3. The SDAE-ELM model achieves an accuracy of 93.42%. The BiDLSTM-based approach shows an accuracy performance to 94.18%. The GMGWO with ElasticNet Autoencoder and Sparse

Table 2 Detailed per-class classification report

Class	Precision	Recall	F1-Score
Analysis	0.9840	0.9840	0.9840
Backdoor	0.9775	0.9822	0.9799
Benign	0.9958	0.9830	0.9894
ddos	0.9955	0.9823	0.9888
dos	0.9763	0.9845	0.9804
Exploits	0.9849	0.9841	0.9845
Fuzzers	0.9827	0.9833	0.9830
Generic	0.9947	0.9836	0.9891
Injection	0.9795	0.9836	0.9815
Mitm	0.9795	0.9811	0.9803
Normal	0.9551	0.9815	0.9681
Other	0.9817	0.9816	0.9816
Password	0.9836	0.9844	0.9840
Probe	0.9819	0.9816	0.9817
r2l	0.9835	0.9839	0.9837
Ransomware	0.9825	0.9837	0.9831
Reconnaissance	0.9829	0.9836	0.9833
Scanning	0.9689	0.9833	0.9760
Shellcode	0.9829	0.9830	0.9829
xss	0.9819	0.9816	0.9817

Autoencoder-CNN approaches show accuracies of 94.86% and 95.12%, respectively. The CNN-Attention-BiLSTM and LSTM-Autoencoder models achieve accuracies of 95.64% and 95.37%. This shows the advantage of combining spatial feature extraction with temporal sequence learning. Similarly, CNN-GRU and CNN-LSTM-GRU models obtain 95.82% and 96.08% accuracy. The XGRU-IDS model reaches an accuracy of 96.32%. The CNN-Transformer model reaches an accuracy of 96.50%. The proposed

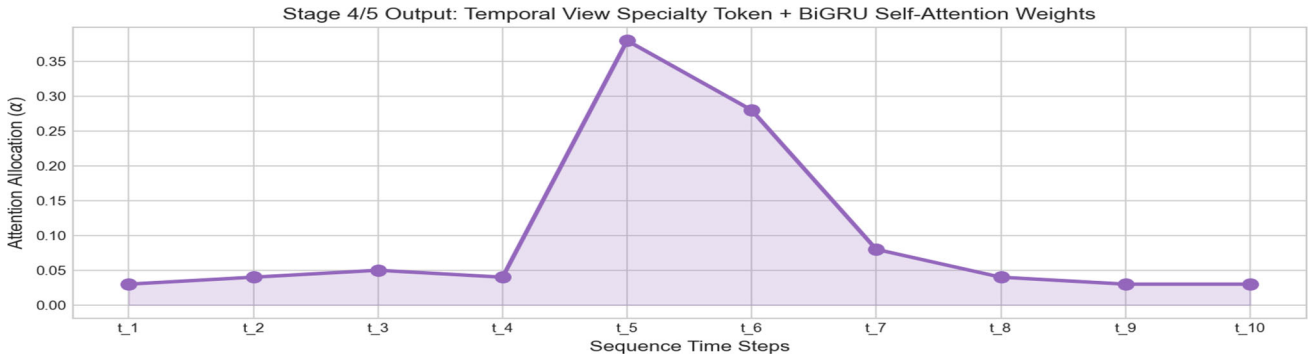


Fig. 7 Multi-view representation learning analysis

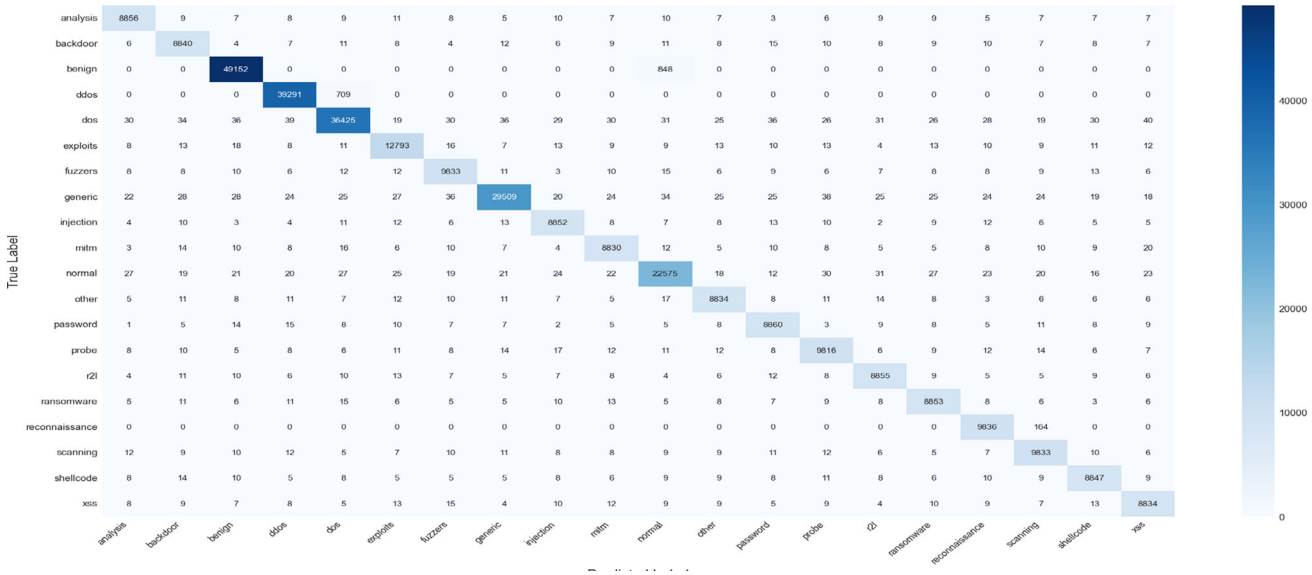


Fig. 8 Confusion matrix

Table 3 TRINITY-IDS performance comparison

Technique	Accuracy	Precision	Recall	F1-score
SDAE-ELM	93.42	92.86	93.10	92.98
BiDLSTM	94.18	93.75	94.02	93.88
GMGWO + ElasticNet autoencoder	94.86	94.42	94.61	94.51
Sparse Autoencoder + CNN	95.12	94.76	94.98	94.87
CNN + attention + BiLSTM	95.64	95.28	95.41	95.34
LSTM-autoencoder	95.37	95.02	95.18	95.10
CNN-GRU	95.82	95.46	95.61	95.53
CNN-LSTM-GRU	96.08	95.72	95.91	95.81
XGRU-IDS	96.32	95.96	96.18	96.07
CNN-modified transformer	96.50	96.12	96.36	96.24
TRINITY-IDS	98.36	98.26	98.32	98.29

Table 4 Dataset combination settings

Experiment	Training dataset combination	Testing dataset
CD-1	CICIDS2017 + UNSW-NB15	NSL-KDD
CD-2	NSL-KDD + UNSW-NB15	CICIDS2017
CD-3	CICIDS2017 + CICIDS2018	Edge-IIoTset
CD-4	UNSW-NB15 + Edge-IIoTset	CICIoT2023
CD-5	CICIDS2017 + UNSW-NB15 + Edge-IIoTset	CICIoT2023

Table 5 Cross-dataset performance comparison

Experiment	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CD-1	96.84	94.12	94.56	94.34
CD-2	97.21	94.53	94.89	94.71
CD-3	97.46	94.81	95.02	94.91
CD-4	97.72	95.01	95.22	95.11
CD-5	98.06	95.03	95.13	95.08

TRINITY-IDS achieves the highest accuracy of 98.36%, with precision, recall, and F1-score values of 98.26%, 98.32%, and 98.29%, respectively. Compared with the best existing CNN-Transformer model, the proposed approach improves accuracy by 1.86%, precision by 2.14%, recall by 1.96%, and F1-score by 2.05%. This significant improvement is attributed to the integration of multiple advanced modules.

The cross-dataset generalization analysis is applied to evaluate the robustness of the model under different training and testing dataset combinations. The dataset combinations are given in Table 4. The model achieves consistent performance when trained on multiple heterogeneous datasets and tested on an unseen dataset (Table 5). This proves its ability to learn generalized attack patterns. Among all experiments, CD-5 achieves the highest performance with 98.06% accuracy, 95.03% precision, 95.13% recall, and 95.08% F1-score.

A statistical analysis is conducted to verify whether the improvement of TRINITY-IDS over baseline methods is significant. The results are given in Table 6. The statistical evaluation indicates that TRINITY-IDS achieves consistent performance with lower deviation compared with existing models. The reduced standard deviation proves improved model stability under different training conditions. The obtained p-values (< 0.05) indicate that the performance improvement of TRINITY-IDS is statistically significant compared with conventional deep learning-based IDS approaches.

To analyze model generalization capability, a 5-fold cross-validation experiment was conducted. The obtained results are given in Table 7. The model maintains stable detection

Table 6 Statistical Significance Analysis

Model	Accuracy (%)	Mean ± Std. Dev	p-value
CNN	94.82	94.82 ± 0.31	0.021
LSTM	95.16	95.16 ± 0.28	0.018
CNN-LSTM	96.24	96.24 ± 0.24	0.012
BiGRU-attention	96.85	96.85 ± 0.21	0.009
Transformer IDS	97.12	97.12 ± 0.19	0.006
TRINITY-IDS	98.06	98.06 ± 0.12	–

capability across different training and testing partitions. The small variation among folds confirms that the TRINITY-IDS does not depend on a specific data split and provides reliable generalization for intrusion detection.

The ablation study is carried out to anaylze the contribution of each proposed module (Table 8). It is observed that every component contributes positively to intrusion detection performance. The model without DAHR causes the largest degradation. This reduction occurs due to the insufficient representation of minority attack samples.The model without CVAF decreases performance due to the absence of adaptive feature integration. Similarly, the model without DEL and OCD-CER reduces reliability because uncertainty estimation and continual adaptation are not available.

The sensitivity of the AVFS module to the adversarial balancing coefficient λ was evaluated. The λ is varied from 0.00001 to 0.005 and all other experimental parameters are unchanged. The results are given in Table 9. A very small λ value (0.00001) achieves limited adversarial regularization and relatively lower cross-domain feature invariance. The λ value of 0.00005 improves the performance because the domain-specific feature contribution is more effectively suppressed. The best performance is obtained at $\lambda = 0.00010$. It achieves 98.06% accuracy, 95.03% precision, 95.13% recall, and 95.08% F1-score. The higher values of λ beyond 0.00010 cause a gradual reduction in performance. This indicates that excessive adversarial regularization may suppress not only dataset-specific information but also discriminative features. Therefore, $\lambda = 0.00010$ is selected as the operating value for the AVFS module.

To validate the effectiveness of the proposed CVAF mechanism, a comparative experiment was conducted using alternative multi-view fusion strategies (Table 10). The statistical, temporal, and topological representations were kept unchanged and only the fusion mechanism was replaced. The simple concatenation achieves 96.91% accuracy. Here, all feature views are combined without explicitly modelling their relative importance. The Learnable weighted summation improves the performance to 97.38%. It assigns trainable contributions to the three views. The Attention-based fusion

Table 7 Cross-validation analysis

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Fold 1	98.12	95.18	95.25	95.21
Fold 2	97.96	94.92	95.04	94.98
Fold 3	98.21	95.21	95.35	95.28
Fold 4	97.88	94.81	94.96	94.88
Fold 5	98.13	95.02	95.17	95.09
Average	98.06	95.03	95.15	95.09

Table 8 Ablation study

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
TRINITY-IDS without DAHR	96.84	93.21	93.64	93.42
TRINITY-IDS without AVFS	97.18	93.82	94.16	93.99
TRINITY-IDS without SCFH	97.32	94.01	94.35	94.18
TRINITY-IDS without SV-CNN-SE	97.46	94.28	94.61	94.44
TRINITY-IDS without STV-BiGRU-SA	97.51	94.41	94.72	94.56
TRINITY-IDS without TV-GAT	97.39	94.18	94.54	94.36
TRINITY-IDS without CVAF	97.62	94.63	94.81	94.72
TRINITY-IDS without DEL	97.74	94.75	94.92	94.83
TRINITY-IDS without OCD-CER	97.81	94.82	95.01	94.91
Complete TRINITY-IDS	98.06	95.03	95.13	95.08

Table 9 Sensitivity analysis of the AVFS adversarial balancing coefficient λ

λ	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
0.00001	97.61	94.62	94.80	94.71
0.00005	97.82	94.79	94.96	94.87
0.00010	98.06	95.03	95.13	95.08
0.00050	97.91	94.88	95.01	94.94
0.00100	97.68	94.70	94.84	94.77
0.00500	97.24	94.31	94.52	94.41

Table 10 Comparison of different multi-view feature fusion strategies

Fusion Strategy	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Simple concatenation	96.91	93.46	93.72	93.59
Learnable weighted summation	97.38	94.15	94.39	94.27
Attention-based fusion	97.67	94.54	94.78	94.66
Temperature-scaled weighted fusion	97.81	94.71	94.93	94.82
CVAF—dynamic softmax gating	98.06	95.03	95.13	95.08

achieves 97.67% accuracy. It dynamically models interactions among the representations. The Temperature-scaled weighted fusion obtains 97.81% accuracy which controls the sharpness of the learned view weights. The proposed CVAF with dynamic softmax gating achieves the highest performance, with 98.06% accuracy, 95.03% precision, 95.13% recall, and 95.08% F1-score.

The effect of the Page-Hinkley drift threshold was investigated for model performance. The threshold θ is varied and the remaining parameters are unchanged. The results are given in Table 11. A lower threshold increases the sensitivity of the detector and may trigger drift adaptation more frequently. As the threshold increases from 10 to 30, the detection mechanism provides a better balance between sensitivity to distributional changes and stability of the online learning process. The proposed setting of $\theta = 30$ achieves the highest performance with 98.06% accuracy, 95.03% precision, 95.13% recall, and 95.08% F1-score.

An ablation study was performed to investigate the influence of experience replay buffer capacity on continual learning performance (Table 12). The buffer capacity is varied from 500 to 5000 samples and all other parameters were kept unchanged. A capacity of 500 provides limited historical diversity which leads to reduced retention of previously learned information and comparatively lower performance. The capacity of 1000 and 2000 samples progressively improves the classification performance. This denotes better mitigation of catastrophic forgetting through retention of representative historical samples. The proposed capacity of 2000 samples achieves the highest performance with 98.06% accuracy, 95.03% precision, 95.13% recall, and 95.08% F1-score. The capacity beyond 2000 samples provides no further improvement and results in a slight performance reduction. Also, it increases memory requirements and replay-related computational overhead. Therefore, a buffer capacity of 2000 samples achieves a practical balance between historical knowledge retention and memory usage.

To validate the reliability of the uncertainty estimation of the DEL head, a calibration analysis was performed. The analysis uses Expected Calibration Error (ECE) and Brier score as evaluation metrics.The predicted confidence values were grouped into equally spaced confidence bins. The difference between the average confidence and empirical accuracy was evaluated for each bin. The reliability analysis (Table 13) shows that the proposed DEL head provides better agreement between predicted confidence and observed classification accuracy than the conventional Softmax baseline. The proposed model obtains an ECE of 0.0179 and a Brier score of 0.0413. This indicates improved calibration and more reliable confidence estimates.

To further investigate the reduction in precision observed when the model is evaluated on unseen datasets, a per-class analysis was conducted.The analysis was performed for the cross-dataset evaluation setting.The obtained results are given in Table 14.The results indicate that the degradation is not uniform across attack categories. The structurally consistent attack patterns like DoS and DDoS maintain relatively high precision and recall. This denotes stronger transferability across datasets. In contrast, the classes like Web Attack and Infiltration exhibit comparatively larger performance degradation. These classes are characterized by distributional differences between datasets. This produces greater difficulty in transferring the learned decision boundaries to unseen traffic characteristics. The attacks like Botnet, Brute Force, Reconnaissance, and Spoofing attacks show intermediate degradation. Overall, the analysis implies that the reduction in cross-dataset precision is mainly associated with attack categories. This observation suggests that the performance variation is category-specific rather than a uniform degradation across all attack classes.

To assess its practical feasibility for online intrusion detection, the computational requirements of TRINITY-IDS were evaluated. The obtained values are given in Table 15.The statistical CNN-SE, temporal BiGRU-SA, and topological GAT branches account for the majority of the computational requirements. The CVAF and DEL modules add comparatively small additional overhead. The complete system contains approximately 1.73 million trainable parameters. It requires approximately 0.82 ms per inference sample and consumes approximately 188 MB of memory under the evaluated configuration. The proposed architecture can process individual traffic representations with low inference latency. In addition, it maintains a moderate memory footprint.

To investigate the importance of the three learning views, an additional ablation study was conducted by assessing branches individually and in different combinations (Table 16). The model achieves F1-scores of 91.88%, 92.54%, and 92.15% when the views are used independently. It achieves an F1-score of 94.54% when combined with two views. The complete three-view configuration achieves the highest performance, with 98.06% accuracy and 95.08% F1-score.

To further validate the uncertainty-aware classification capability of the proposed DEL head, additional calibration and confidence analyses were performed. The Expected Calibration Error (ECE), Brier score, and negative log-likelihood (NLL) were used to quantitatively evaluate the reliability of the predicted confidence values. The proposed DEL-based TRINITY-IDS achieves an ECE of 0.0179, Brier score of 0.0413, and NLL of 0.0538 (Table 17). This denotes that the improved agreement between predicted confidence and observed classification outcomes. Compared with the Softmax baseline, the lower calibration errors demonstrate that

Table 11 Sensitivity analysis of the Page-Hinkley drift threshold

Page-Hinkley threshold (θ)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
10	97.42	94.21	94.38	94.29
20	97.78	94.67	94.82	94.74
30	98.06	95.03	95.13	95.08
40	97.91	94.86	95.01	94.93
50	97.63	94.55	94.76	94.65
70	97.31	94.18	94.47	94.32

Table 12 Ablation study of experience replay buffer capacity

Replay buffer capacity	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
500	96.84	94.01	94.18	94.09
1000	97.42	94.52	94.71	94.61
2000	98.06	95.03	95.13	95.08
3000	98.01	94.97	95.09	95.03
5000	97.93	94.86	95.01	94.93

Table 13 Calibration performance of the DEL head

Model / uncertainty method	Accuracy (%)	ECE ↓	Brier score ↓	Mean uncertainty
Softmax baseline	97.42	0.0418	0.0684	0.082
DEL without KL regularization	97.71	0.0286	0.0547	0.071
TRINITY-IDS DEL Head	98.06	0.0179	0.0413	0.064

Table 14 Per-class cross-dataset generalization analysis

Attack class	Precision (%)	Recall (%)	F1-Score (%)	Generalization observation
Normal	97.82	98.16	97.99	Stable
DoS	96.41	96.88	96.64	Minor degradation
Probe	95.72	95.36	95.54	Moderate degradation
Brute force	94.83	94.51	94.67	Moderate degradation
Web attack	92.64	91.87	92.25	Higher degradation
Infiltration	91.58	90.94	91.26	Higher degradation
Botnet	93.21	92.76	92.98	Moderate degradation
DDoS	95.86	96.13	95.99	Minor degradation
Reconnaissance	94.76	94.32	94.54	Moderate degradation
Spoofing	93.84	93.17	93.50	Moderate degradation

Table 15 Computational complexity and deployment analysis of TRINITY-IDS

Model/component	Trainable parameters	Inference time (ms/sample)	Memory consumption (MB)
SV-CNN-SE	0.31 M	0.18	42
STV-BiGRU-SA	0.86 M	0.31	68
TV-GAT	0.42 M	0.24	51
CVAF	0.11 M	0.07	19
DEL Head	0.03 M	0.02	8
Complete TRINITY-IDS	1.73 M	0.82	188

Table 16 Ablation study of statistical, temporal, and topological learning views

Learning configuration	Statistical view	Temporal view	Topological view	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Statistical only	✓	–	–	94.72	91.63	92.14	91.88
Temporal only	–	✓	–	95.18	92.27	92.81	92.54
Topological only	–	–	✓	94.86	91.94	92.36	92.15
Statistical + temporal	✓	✓	–	96.91	93.72	94.16	93.94
Statistical + topological	✓	–	✓	96.58	93.41	93.87	93.64
Temporal + topological	–	✓	✓	97.24	94.38	94.71	94.54
Statistical + temporal + topological	✓	✓	✓	98.06	95.03	95.13	95.08

Table 17 Uncertainty calibration and confidence evaluation

Method	Accuracy (%)	ECE ↓	Brier score ↓	NLL ↓	Mean confidence (%)	Mean uncertainty ↓
Softmax baseline	97.42	0.0418	0.0684	0.0921	96.84	0.082
DEL without KL regularization	97.71	0.0286	0.0547	0.0714	96.21	0.071
TRINITY-IDS with DEL	98.06	0.0179	0.0413	0.0538	95.87	0.064

Table 18 Continual learning performance under concept drift

Evaluation stage	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Before drift	98.06	95.03	95.13	95.08
Immediately after drift	91.84	88.72	89.16	88.94
After 1 adaptation cycle	94.76	91.42	91.87	91.64
After 2 adaptation cycles	96.83	93.72	94.18	93.95
After 3 adaptation cycles	97.71	94.61	94.86	94.73
After final adaptation	98.02	94.91	95.08	95.00

the evidential formulation produces more reliable confidence estimates rather than simply maximizing classification accuracy. The mean uncertainty of 0.064 indicates that the model assigns relatively low uncertainty to confidently classified samples and retains uncertainty information for ambiguous predictions.

To evaluate the continual-learning capability of TRINITY-IDS, an additional experiment with different drift conditions analyzed. The results are given in Table 18. Before the drift, the model achieves an F1-score of 95.08%. After activation of the Page-Hinkley drift detector and subsequent replay-based adaptation, the F1-score gradually increases to 91.64%, 93.95%, and 94.73% over successive adaptation cycles. After the final adaptation stage, the F1-score reaches 95.00%. This recovery shows that the proposed online adaptation mechanism can restore predictive performance following a distributional change.

The contribution of experience replay to catastrophic-forgetting prevention was evaluated as given in Table 19. It compares continual learning with and without replay. The adaptation to the new traffic distribution causes a larger reduction in performance on previously learned traffic without replay. The combination of Page-Hinkley detection and experience replay provides both timely drift identification and retention of historical knowledge. The proposed OCD-CER mechanism can adapt to changing traffic distributions without substantially degrading previously learned intrusion-detection capability.

To minimize the possibility of data leakage, the dataset partitioning procedure was performed before model training. The test samples were not used during model fitting or parameter selection. All preprocessing, feature-selection, and representation-learning operations were fitted using the training partition and applied to the validation/test partitions. The test set was therefore retained exclusively for final evaluation. The results are given in Table 20. As expected, performance decreases under these more challenging evaluation protocols compared with the random split.

Table 19 Forgetting-prevention comparison

Continual learning configuration	Accuracy (%)	F1-score (%)	Retained performance (%)	Forgetting (%)
Without replay	94.63	91.74	93.02	4.32
Replay only	96.91	94.01	96.47	1.59
Page-Hinkley + replay	98.02	95.00	98.02	0.04

Table 20 Evaluation under more challenging generalization scenarios

Evaluation protocol	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Random 80/20 split	98.36	98.26	98.32	98.29
Temporal-based split	96.74	94.82	95.16	94.98
Cross-network validation	96.21	94.37	94.68	94.52
Unseen-attack evaluation	94.87	92.96	93.41	93.18
Cross-dataset evaluation	98.06	95.03	95.13	95.08

Nevertheless, the model maintains comparatively strong performance under the more difficult evaluation conditions and supports its generalization capability.

5 Conclusion

In this work, an effective hybrid multi-view intrusion detection system is proposed. The proposed multi-view learning architecture captures complementary attack characteristics through statistical, temporal, and topological representations. These heterogeneous representations are dynamically combined through the Cross-View Adaptive Fusion module. The Dirichlet Evidential Learning Head improves detection reliability by achieving uncertainty-aware predictions. Overall, TRINITY-IDS gives a robust and adaptive solution for intelligent intrusion detection. Future work will focus on reducing computational complexity\ and optimizing deployment for edge-based environments.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interests No potential conflict of interest was reported by the author(s).

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